TUTORIAL 15

Completion: Orthogonal Matrices, the Spectral Theorem, and SVD

15.1 Orthogonal Matrices and the Spectral Theorem

DEFINITION 15.1

A real square matrix Q is orthogonal if $Q^TQ = I$. Equivalently, its columns form an orthonormal basis, or $Q^{-1} = Q^T$. In particular, $\|Qx\| = \|x\|$ for the standard inner product. For a complex matrix, $A^* = \overline{A}^T$ denotes the conjugate transpose. A matrix U is unitary if $U^*U = I$. The adjoint map satisfies $\langle Tv, w\rangle = \langle v, T^*w\rangle$ and is represented by A^* in an orthonormal basis.

PROPOSITION 15.1

On a finite-dimensional complex inner product space, every self-adjoint map has an orthonormal eigenbasis, and all its eigenvalues are real. This is the spectral theorem for self-adjoint maps. Thus for every Hermitian matrix $A = A^*$, there are a unitary matrix U and a real diagonal matrix D such that

$$U^*AU = D, \qquad A = UDU^*.$$

Column j of U is an eigenvector of A belonging to d_{jj}. Over $\mathbb{R}$, every real symmetric matrix $A = A^T$ has a real orthonormal eigenbasis; hence there is an orthogonal matrix Q such that

$$Q^TAQ = D, \qquad A = QDQ^T.$$

Diagonalization by an orthogonal matrix is called **orthogonal diagonalization**; over the complex numbers, we use the term **unitary diagonalization**.

RECALL AND TOOLS

To find an orthogonal diagonalization of a real symmetric matrix, find a basis of each eigenspace, apply Gram-Schmidt within each eigenspace as needed, and normalize. Eigenvectors of a symmetric matrix belonging to distinct eigenvalues are already orthogonal. Place the unit vectors in the columns of Q, and place the corresponding eigenvalues in the same order on the diagonal of D. Over the complex numbers, the same construction applies to a Hermitian matrix, using U^* in place of Q^T.

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15.2 Linear Maps on Inner Product Spaces

EXERCISE 15.1

Let $(V, \langle\cdot,\cdot\rangle)$ be a real inner product space of dimension $n \geq 3$. Let $v, w \in V$ be orthonormal: that is, $\langle v, v\rangle = \langle w, w\rangle = 1$ and $\langle v, w\rangle = 0$. Define the linear map $T : V \to V$ by

$$T(u) = \langle u, v\rangle\, w + \langle u, w\rangle\, v.$$

1. Please find all eigenvalues of T, all eigenvectors, and the characteristic polynomial of T.
2. Please prove that $\|T(u)\| \leq \|u\|$ for every $u \in V$, and please find all u for which equality holds.

WORKED SOLUTION

1. Start with the eigenvalue equation and compare components using an orthogonal decomposition:

   $$T(u) = \lambda u.$$

   Decompose each $u \in V$ using the orthogonal direct sum

   $$V = \text{span}\{v, w\} \oplus (\text{span}\{v, w\})^{\perp}.$$

   Here $\{v, w\}^{\perp}$ denotes the orthogonal complement of $\text{span}\{v, w\}$. Thus there are unique $\alpha, \beta \in \mathbb{R}$ and $y \in \{v, w\}^{\perp}$ such that

   $$u = \alpha v + \beta w + y.$$

   Now compute $T(u)$: since $y \perp v, w$, we have $\langle y, v\rangle = \langle y, w\rangle = 0$, so

   $$\langle u, v\rangle = \langle \alpha v + \beta w + y,\ v\rangle = \alpha, \qquad \langle u, w\rangle = \langle \alpha v + \beta w + y,\ w\rangle = \beta.$$

   Hence

   $$T(u) = \alpha w + \beta v.$$

   Substitute into the eigenvalue equation $T(u) = \lambda u$ to obtain

   $$\alpha w + \beta v = \lambda(\alpha v + \beta w + y).$$

   Compare components in the direct sum $\text{span}\{v, w\} \oplus \{v, w\}^{\perp}$. The component in $\{v, w\}^{\perp}$ on the left is zero, so necessarily

   $$\lambda y = 0.$$

   Consequently:

   - If $\lambda \neq 0$, then $y = 0$, so $u \in \text{span}\{v, w\}$.
   - If $\lambda = 0$, then $T(u) = 0$, and $\alpha w + \beta v = 0$ gives $\alpha = \beta = 0$, so $u = y \in \{v, w\}^{\perp}$.

   Now consider $\lambda \neq 0$. Then $u = \alpha v + \beta w$ and $y = 0$. The eigenvalue equation becomes

   $$\alpha w + \beta v = \lambda(\alpha v + \beta w).$$

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EXERCISE 15.1 CONTINUED

Comparing coefficients of v and w gives the system

$$\beta = \lambda\alpha, \qquad \alpha = \lambda\beta.$$

Substitution gives $\alpha = \lambda(\lambda\alpha) = \lambda^2\alpha$, so

$$(\lambda^2 - 1)\alpha = 0.$$

If $\alpha = 0$, then $\beta = \lambda\alpha = 0$ as well, giving $u = 0$, contrary to the definition of an eigenvector. Thus $\alpha \neq 0$, and

$$\lambda^2 - 1 = 0 \quad \Longleftrightarrow \quad \lambda \in \{1, -1\}.$$

Now determine the eigenvectors:

- For $\lambda = 1$, $\beta = \alpha$, so $u = \alpha(v + w)$; hence

$$V_1 = \text{span}\{v + w\}.$$

- For $\lambda = -1$, $\beta = -\alpha$, so $u = \alpha(v - w)$; hence

$$V_{-1} = \text{span}\{v - w\}.$$

- For $\lambda = 0$, we obtained $u \in \{v, w\}^\perp$, so

$$V_0 = \{v, w\}^\perp.$$

In particular, there are no other eigenvalues, because the eigenvalue equation forced $\lambda \in \{0, 1, -1\}$.

To compute the characteristic polynomial, choose an orthonormal basis of V of the form

$$B = (v, w, e_3, \ldots, e_n),$$

where $e_3, \ldots, e_n$ is an orthonormal basis of $\{v, w\}^\perp$. Then

$$T(v) = w, \qquad T(w) = v, \qquad T(e_i) = 0 \ \ (i = 3, \ldots, n).$$

The matrix representing T relative to B is therefore

$$[T]_B = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

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EXERCISE 15.1 CONTINUED

Hence its characteristic polynomial is

$$p_T(\lambda) = \det(\lambda I - [T]_B) = \det \begin{pmatrix} \lambda & -1 & 0 & \cdots & 0 \\ -1 & \lambda & 0 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda \end{pmatrix} = (\lambda^2 - 1)\lambda^{n-2}.$$

2. Set $\alpha := \langle u, v \rangle$ and $\beta := \langle u, w \rangle$, and define

$$s := \alpha v + \beta w \in \text{span}\{v, w\}.$$

Since v, w are orthonormal,

$$\|s\|^2 = \langle s, s \rangle = \alpha^2 + \beta^2 = \langle u, v \rangle^2 + \langle u, w \rangle^2 = \|T(u)\|^2.$$

Also,

$$\langle u, s \rangle = \alpha \langle u, v \rangle + \beta \langle u, w \rangle = \alpha^2 + \beta^2 = \|s\|^2.$$

By the Cauchy-Schwarz inequality,

$$|\langle u, s \rangle| \leq \|u\| \, \|s\|.$$

Since $|\langle u, s \rangle| = \|s\|^2$, we obtain

$$\|s\|^2 \leq \|u\| \, \|s\|.$$

If $s = 0$, then $\|T(u)\| = \|s\| = 0 \leq \|u\|$. If $s \neq 0$, divide by $\|s\|$ to obtain

$$\|T(u)\| = \|s\| \leq \|u\|.$$

To check equality, write $u = s + y$ with $y \in \{v, w\}^\perp$. By the Pythagorean identity,

$$\|u\|^2 = \|s\|^2 + \|y\|^2 = \|T(u)\|^2 + \|y\|^2.$$

Thus $\|T(u)\| = \|u\|$ if and only if $y = 0$, or equivalently $u \in \text{span}\{v, w\}$. This argument also includes $s = 0$, in which case equality requires $u = 0$.

EXERCISE 15.2

Let $n \geq 3$ be a natural number and let $a \in \mathbb{R}$. Define $A_n(a) \in M_n(\mathbb{R})$ by

$$(A_n(a))_{ij} = \begin{cases} 1, & i = j, \\ a, & i \neq j. \end{cases}$$

Recall the definition: a Hermitian matrix $H \in M_n(\mathbb{C})$ is positive semidefinite if $x^* H x \geq 0$ for every $x \in \mathbb{C}^n$, and positive definite if $x^* H x > 0$ for every $x \neq 0$.

1. For every a, please find all eigenvalues of $A_n(a)$ and their multiplicities.
2. Please determine for which values of a the matrix $A_n(a)$ is positive definite, and for which values it is positive semidefinite.

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EXERCISE 15.2 CONTINUED

3. Please find $\operatorname{rank}(A_n(a))$ and $\ker(A_n(a))$ in the two boundary cases where $A_n(a)$ is positive semidefinite but not positive definite.

In addition, let $B \in M_n(\mathbb{C})$ and please write $B^* = \overline{B}^T$. Define

$$H := B^*B + I.$$

(4) Please prove that H is positive definite and hence invertible.

(5) Please prove that, for every $x \in \mathbb{C}^n$,

$$x^*H^{-1}x \leq \|x\|^2,$$

and also prove that

$$\det(H) \geq 1,$$

with equality if and only if $B = 0$.

WORKED SOLUTION

Write $e = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$. We have the decomposition

$$A_n(a) = (1-a)I + a\,ee^T,$$

because every entry of ee^T equals 1.

1. Note that $ee^Te = (e^Te)\,e = n\,e$, so

$$A_n(a)e = (1-a)e + a\,ee^Te = (1-a)e + an\,e = \big(1 + (n-1)a\big)e,$$

Thus e is an eigenvector with eigenvalue $\lambda_1 = 1 + (n-1)a$.
Now let $x \in \mathbb{R}^n$ satisfy $e^Tx = 0$. Then $ee^Tx = e(e^Tx) = 0$, so

$$A_n(a)x = (1-a)x + a \cdot 0 = (1-a)x.$$

Consequently, every $x \in e^\perp \setminus \{0\}$ is an eigenvector with eigenvalue $\lambda_2 = 1 - a$. Since $\mathbb{R}^n = \operatorname{span}\{e\} \oplus e^\perp$, these are all eigenvalues. If $a \neq 0$, then $\lambda_1 \neq \lambda_2$, so

$$\begin{aligned} &\lambda_1 = 1 + (n-1)a && \text{with algebraic and geometric multiplicity } 1, \\ &\lambda_2 = 1 - a && \text{with algebraic and geometric multiplicity } n-1. \end{aligned}$$

If $a = 0$, then $A_n(0) = I$, so its only eigenvalue is 1, with algebraic and geometric multiplicity n.

2. The matrix $A_n(a)$ is real symmetric and hence Hermitian. It is therefore positive definite exactly when all its eigenvalues are strictly positive, and positive semidefinite exactly when all its eigenvalues are nonnegative. Thus:

$$A_n(a) \succ 0 \iff \begin{cases} 1-a > 0, \\ 1 + (n-1)a > 0, \end{cases} \iff -\frac{1}{n-1} < a < 1,$$

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EXERCISE 15.2 CONTINUED

and

$$A_n(a) \succeq 0 \iff \begin{cases} 1-a \geq 0, \\ 1+(n-1)a \geq 0, \end{cases} \iff -\frac{1}{n-1} \leq a \leq 1.$$

3. If $a = 1$, then $\lambda_2 = 0$ has multiplicity $n-1$, and $\lambda_1 = n$. Hence

$$\operatorname{rank}(A_n(1)) = 1, \qquad \ker(A_n(1)) = e^{\perp} = \{x \in \mathbb{R}^n : \sum_{i=1}^{n} x_i = 0\}.$$

If $a = -\frac{1}{n-1}$, then $\lambda_1 = 0$ has multiplicity 1, and $\lambda_2 = 1 - a = \frac{n}{n-1} > 0$ has multiplicity $n-1$. Hence

$$\operatorname{rank}\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = n-1, \qquad \ker\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = \operatorname{span}\{e\}.$$

Now turn to $H = B^*B + I$.

(4) Let $x \in \mathbb{C}^n$. Then

$$x^*Hx = x^*(B^*B)x + x^*Ix = \|Bx\|^2 + \|x\|^2.$$

Hence $x^*Hx \geq 0$ for every x, and $x^*Hx > 0$ for every $x \neq 0$, since $\|x\|^2 > 0$. Thus $H \succ 0$, so H is invertible.

(5) From $H = I + B^*B$, we obtain $H - I = B^*B \succeq 0$; thus $x^*Hx \geq x^*Ix = \|x\|^2$ for every x. Let $\lambda_{\min}$ be the smallest eigenvalue of H. The preceding inequality gives $\lambda_{\min} \geq 1$. Every eigenvalue of H^{-1} is therefore at most 1, so for every x,

$$x^*H^{-1}x \leq \|x\|^2.$$

For the determinant inequality, write the eigenvalues of H as $\lambda_1, \ldots, \lambda_n$. Since $H \succ 0$, we have $\lambda_i > 0$; since $\lambda_{\min} \geq 1$, in fact $\lambda_i \geq 1$ for every i. Hence

$$\det(H) = \prod_{i=1}^{n} \lambda_i \geq 1.$$

Equality holds if and only if $\lambda_i = 1$ for every i: equivalently, $H = I$, or $B^*B = 0$. Finally, $B^*B = 0$ implies $\|Bx\|^2 = x^*B^*Bx = 0$ for every x. Thus $Bx = 0$ for all x, so $B = 0$.

EXERCISE 15.3 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be symmetric matrices.

1. Please prove that there is an orthogonal matrix Q for which both Q^TAQ and Q^TBQ are diagonal if and only if $AB = BA$.

2. Please find such a matrix for

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 6 \end{pmatrix},$$

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EXERCISE 15.3 CHALLENGE CONTINUED

and also give the diagonal matrix representing AB in the same basis.

WORKED SOLUTION

1. If $Q^TAQ = D_A$ and $Q^TBQ = D_B$ are diagonal, then $A = QD_AQ^T$ and $B = QD_BQ^T$. Since diagonal matrices commute,

$$AB = QD_AD_BQ^T = QD_BD_AQ^T = BA.$$

Conversely, suppose $AB = BA$. By the spectral theorem, $\mathbb{R}^n$ is the orthogonal direct sum of the eigenspaces of A. If $x \in E_\lambda(A)$, then

$$A(Bx) = B(Ax) = B(\lambda x) = \lambda Bx,$$

so $Bx \in E_\lambda(A)$. Thus B preserves every eigenspace of A. On each such space, the restriction of B is self-adjoint for the induced inner product: for all x, y in that space,

$$\langle Bx, y\rangle = \langle x, By\rangle,$$

because $B^T = B$. By the spectral theorem on that space, choose an orthonormal basis of eigenvectors of B. Each vector is also an eigenvector of A belonging to λ. Combining these bases over all eigenvalues of A gives an orthonormal basis of $\mathbb{R}^n$ common to both matrices. Placing it in the columns of Q gives the required diagonalizations, including the case of repeated eigenvalues.

2. Choose

$$q_1 = \frac{1}{\sqrt{2}}(1,1,0)^T, \qquad q_2 = \frac{1}{\sqrt{2}}(1,-1,0)^T, \qquad q_3 = (0,0,1)^T.$$

Their norms are 1 and their pairwise inner products are 0. Furthermore,

$$Aq_1 = -q_1, \quad Aq_2 = -q_2, \quad Aq_3 = 4q_3, \qquad Bq_1 = 7q_1, \quad Bq_2 = 3q_2, \quad Bq_3 = 6q_3.$$

Thus, for

$$Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

we obtain

$$Q^TAQ = \operatorname{diag}(-1,-1,4), \qquad Q^TBQ = \operatorname{diag}(7,3,6),$$
$$Q^TABQ = \operatorname{diag}(-7,-3,24).$$

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EXERCISE 15.4 CHALLENGE

Let $A \in M_n(\mathbb{R})$ be symmetric and positive definite, meaning $x^T Ax > 0$ for every $x \neq 0$. Please prove that there is a unique symmetric positive definite matrix B satisfying $B^2 = A$.

WORKED SOLUTION

The spectral theorem gives an orthogonal matrix Q with

$$A = QDQ^T, \qquad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Every λ_j is positive, because a corresponding unit eigenvector q_j satisfies $\lambda_j = q_j^T Aq_j > 0$. Define

$$B = Q \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})Q^T.$$

It is symmetric and satisfies $B^2 = A$. For $x \neq 0$, put $y = Q^T x \neq 0$. Then $x^T Bx = \sum_j \sqrt{\lambda_j} y_j^2 > 0$. For uniqueness, let C be another symmetric positive definite square root. Since $CA = C^3 = AC$, it preserves every eigenspace $E_\lambda(A)$. Its restriction to that eigenspace is symmetric and positive definite, so it has an orthonormal eigenbasis. For a vector with $Cv = \mu v$, positivity gives $\mu > 0$, and

$$\mu^2 v = C^2 v = Av = \lambda v.$$

Thus $\mu = \sqrt{\lambda}$. On each eigenspace of A, both B and C act as multiplication by $\sqrt{\lambda}$. These eigenspaces span $\mathbb{R}^n$, proving $C = B$.

15.3 Singular Value Decomposition

NOTATION AND CONVENTIONS

In this part, $A \in M_{m\times n}(\mathbb{R})$ with $m, n \geq 1$, and all inner products are standard. A square Q is orthogonal if $Q^T Q = QQ^T = I$. Its columns form an orthonormal basis.

DEFINITION 15.2

A **singular value decomposition**, abbreviated SVD, is a factorization

$$A = U\Sigma V^T,$$

where $U \in M_m(\mathbb{R})$ and $V \in M_n(\mathbb{R})$ are orthogonal, and $\Sigma \in M_{m\times n}(\mathbb{R})$ has numbers

$$\sigma_1 \geq \cdots \geq \sigma_r > 0$$

in positions $(1,1), \ldots, (r,r)$ and zero elsewhere. Here $r = \operatorname{rank} A$; if $A = 0$, the positive list is empty and $\Sigma = 0$. The numbers σ_i are the **positive singular values**. When including zeros, we list $\min\{m, n\}$ singular values, with their multiplicities.

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RECALL AND TOOLS

Write the columns of U as $u_1, \dots, u_m$, and those of V as $v_1, \dots, v_n$. The factorization gives

$$Av_i = \sigma_i u_i \quad (1 \le i \le r), \qquad Av_i = 0 \quad (r < i \le n),$$

so

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$

For example, $x = \sum_{i=1}^{n} c_i v_i$ is mapped to

$$Ax = \sum_{i=1}^{r} \sigma_i c_i u_i.$$

The basis V describes directions in the domain, and U describes directions in the codomain. Thus $U^T AV = \Sigma$. These are two separate basis choices, even when A is square; similarity uses the same basis change on both sides.

PROPOSITION 15.2

Every real matrix admits an SVD. Indeed, $A^T A$ is symmetric, and for every $x \in \mathbb{R}^n$,

$$x^T A^T Ax = (Ax)^T (Ax) = \|Ax\|^2 \ge 0.$$

The spectral theorem gives an orthonormal eigenbasis v_i with nonnegative eigenvalues λ_i. Order the positive eigenvalues first. For each $\lambda_i > 0$, set

$$\sigma_i = \sqrt{\lambda_i}, \qquad u_i = \frac{Av_i}{\sigma_i}.$$

If $\lambda_i, \lambda_j > 0$, then

$$u_i^T u_j = \frac{v_i^T A^T Av_j}{\sigma_i \sigma_j} = \frac{\lambda_j v_i^T v_j}{\sigma_i \sigma_j} = \begin{cases} 1 & i = j, \\ 0 & i \ne j. \end{cases}$$

If $\lambda_i = 0$, then $\|Av_i\|^2 = v_i^T A^T Av_i = 0$, so $Av_i = 0$. Thus $Av_1, \dots, Av_r$ are linearly independent and span $\operatorname{Im} A$, so there are exactly $r = \operatorname{rank} A \le m$ positive eigenvalues. Extend $u_1, \dots, u_r$ to an orthonormal basis of $\mathbb{R}^m$. The corresponding matrices U, V satisfy $AV = U\Sigma$, hence $A = U\Sigma V^T$.

The positive singular values are therefore the positive square roots of the positive eigenvalues of $A^T A$. The matrix $A^T A$ also has $n - r$ zero eigenvalues; if $n > m$, some of these are absent from the shorter list of $\min\{m, n\}$ singular values.

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EXAMPLE 15.1

Consider the rectangular matrix

$$A = \begin{pmatrix} 2 & 4 \\ -1 & -2 \\ 2 & 4 \end{pmatrix}.$$

The second column is twice the first, so its rank is 1. Also,

$$A^T A = \begin{pmatrix} 4+1+4 & 8+2+8 \\ 8+2+8 & 16+4+16 \end{pmatrix} = \begin{pmatrix} 9 & 18 \\ 18 & 36 \end{pmatrix}.$$

Hence

$$\det(tI - A^T A) = (t-9)(t-36) - 324 = t(t-45).$$

Its singular values are $3\sqrt{5}, 0$. Choose

$$v_1 = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad v_2 = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

Direct multiplication gives

$$Av_1 = \frac{1}{\sqrt{5}}(10, -5, 10)^T = 3\sqrt{5}\,\frac{1}{3}(2, -1, 2)^T, \qquad Av_2 = 0.$$

An orthonormal basis of the codomain is

$$u_1 = \frac{1}{3}(2, -1, 2)^T, \quad u_2 = \frac{1}{\sqrt{5}}(1, 2, 0)^T, \quad u_3 = \frac{1}{3\sqrt{5}}(-4, 2, 5)^T.$$

The numerator dot products are $2 - 2 = 0$, $-8 - 2 + 10 = 0$, and $-4 + 4 = 0$, and their norms are $3, \sqrt{5}, 3\sqrt{5}$. For $U = (u_1\ u_2\ u_3)$ and $V = (v_1\ v_2)$,

$$\Sigma = \begin{pmatrix} 3\sqrt{5} & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad U\Sigma V^T = 3\sqrt{5}\,u_1 v_1^T = \begin{pmatrix} 2 & 4 \\ -1 & -2 \\ 2 & 4 \end{pmatrix}.$$

EXERCISE 15.5

Let $A = U\Sigma V^T \in M_{m\times n}(\mathbb{R})$ be an SVD of rank r. Define

$$B = \sum_{i=1}^{r} \frac{1}{\sigma_i} v_i u_i^T \in M_{n\times m}(\mathbb{R}).$$

For every $b \in \operatorname{Im} A$, please prove that Bb is the unique solution of $Ax = b$ with minimum norm. Please prove that B does not depend on the chosen decomposition.

WORKED SOLUTION

The vectors $u_1, \ldots, u_r$ form an orthonormal basis of the image. Thus, for $b \in \operatorname{Im} A$,

$$b = \sum_{i=1}^{r} \beta_i u_i, \qquad \beta_i = u_i^T b.$$

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EXERCISE 15.5 CONTINUED

Writing $x = \sum_{i=1}^{n} c_i v_i$ gives $Ax = \sum_{i=1}^{r} \sigma_i c_i u_i$. The equation $Ax = b$ is equivalent to $c_i = \beta_i/\sigma_i$ for $i \leq r$, with the other coefficients free. Every solution therefore has the form

$$x = \sum_{i=1}^{r} \frac{\beta_i}{\sigma_i} v_i + \sum_{i=r+1}^{n} c_i v_i = Bb + z, \qquad z \in \ker A.$$

The two sums are orthogonal, so

$$\|x\|^2 = \sum_{i=1}^{r} \frac{\beta_i^2}{\sigma_i^2} + \sum_{i=r+1}^{n} c_i^2 = \|Bb\|^2 + \|z\|^2.$$

The minimum occurs exactly when every $c_i = 0$ for $i > r$, or $z = 0$. Hence Bb is the unique minimum-norm solution. Now take an arbitrary $b \in \mathbb{R}^m$. There is a unique orthogonal decomposition $b = b_{\parallel} + b_{\perp}$, with $b_{\parallel} \in \operatorname{Im} A$ and $b_{\perp} \in (\operatorname{Im} A)^{\perp}$. The formula for B gives $Bb_{\perp} = 0$. Thus $Bb = Bb_{\parallel}$ is the unique minimum-norm solution of $Ax = b_{\parallel}$. Both the projection $b_{\parallel}$ and the unique solution are determined by A, b, so every SVD gives the same Bb. This holds for every b, proving that B itself is independent of the decomposition. The matrix B is called the Moore-Penrose pseudoinverse and is denoted by A^{+}. When $A = 0$, the sum is empty and $A^{+} = 0$, as required.

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EXERCISE 15.6

Let $A \in M_n(\mathbb{R})$ satisfy $A^T = -A$. Please prove that each positive singular value of A has even multiplicity. Please deduce that rank A is even.

WORKED SOLUTION

We have $A^T A = -A^2$. Let $\sigma > 0$ be a singular value, and set

$$E = \ker(A^T A - \sigma^2 I), \qquad d = \dim E.$$

The dimension d is the multiplicity of σ: the symmetric matrix $A^T A$ is diagonalizable, so the multiplicity of σ^2 equals its eigenspace dimension. For $x \in E$, we have $-A^2 x = \sigma^2 x$. Moreover,

$$-A^2(Ax) = A(-A^2 x) = \sigma^2 Ax,$$

so $Ax \in E$. Define $Jx = Ax/\sigma$ on E. Then

$$J^2 x = \frac{A^2 x}{\sigma^2} = -x.$$

In any basis of E, the real matrix $M = [J]$ satisfies $M^2 = -I_d$. Taking determinants yields

$$(\det M)^2 = \det(-I_d) = (-1)^d.$$

The left side is nonnegative, so d must be even. Every positive singular value therefore occurs an even number of times. The rank is the number of positive singular values counted with multiplicities, so it is a sum of even integers and is even.

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